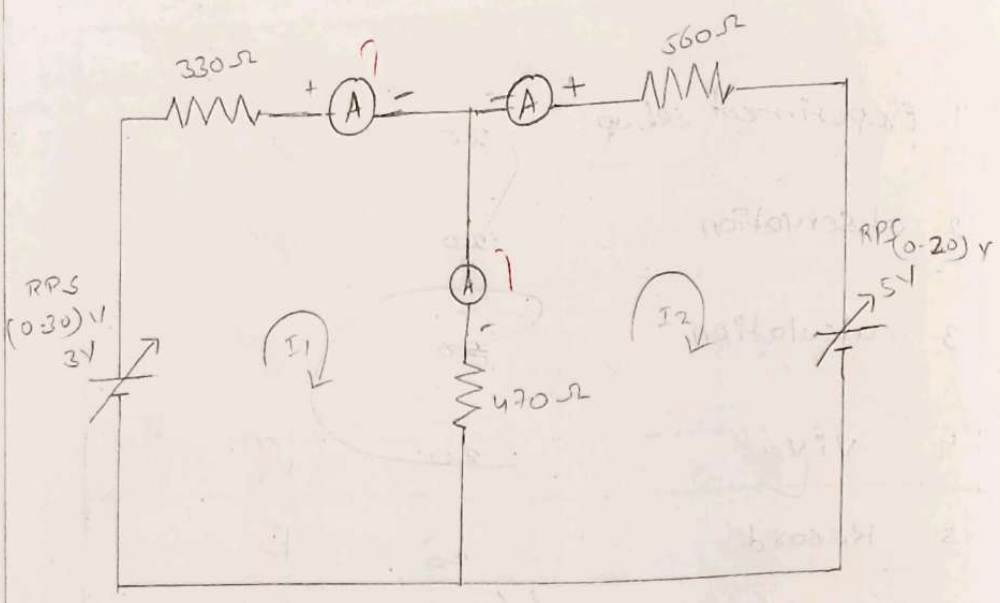
Circuit Diagram :-



Ex.No :- 02

DETERMINATION OF CURRENT IN CIRCUIT USING MASH ANALYSIS

Date: 31/07/26

AIM :-

To determine the current in circuit using mesh analysis both theoretically and practically for a given DC circuit.

APPARATUS REQUIRED :-

S.No	APPARATUS	SPECIFICATION	QUANTITY
1	Regulated power supply (RPS)	(0-30V)	1
2	Ammeter	(0-10mA) MC	3
3	Resistors	330Ω, 470Ω, 560Ω	each 1
4	Bread board	-	1

THEORY :-

Multi-source DC circuits may be analyzed using a mesh current technique. The process involves identifying minimum number of small loops such that every component exist in at least one loop. KVL is then applied to each loop. The loop currents are referred to as mesh currents as each current interlocks or meshes with

TABULAR COLUMN :

Parameters	Theoretical	Practical
I_1	1.22mA	1.24mA
I_2	4.29mA	4.17mA
$I_1 - I_2$	5.51mA	5.44mA

CALCULATION :-

$$\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$R_{11} = 330\,\Omega + 470\,\Omega$

$R_{11} = 800\,\Omega$

$R_{21} = -470\,\Omega$

$R_{12} = -470\,\Omega$

$R_{12} = -470\,\Omega$

$R_{22} = 470 + 560$

$= 1030\,\Omega$

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \cdot \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

$\Delta = 800(1030) - (-470)^2$

$\Delta = 824000 - 220900$

$\Delta = 603100$

$$\Delta_1 = \begin{bmatrix} 3 & -470 \\ -5 & 1030 \end{bmatrix}$$

$= 3(1030) - (-470 \times -5)$

$= 3090 - 2350 \Rightarrow 740$

the surrounding loop currents. As a result there will be a set of simultaneous equations created, an unknown mesh current for each loop. once mesh currents are determined, various branch currents and component voltages may be derived

PROCEDURE:-

1) Give connections as per circuit diagram.
2) switch on the supply. vary the RPS & set a particular input voltage.
3) Note down the readings of ammeters and voltmeters and tabulate them.
4) Reduce the RPS to it's minimum value and switch OFF the supply.
5) ~~Reduce the RPS to it's mini~~ vary the RPS for different input voltages and note down the reading of all meters.
6) using the tabulated values. verify kirchoff's laws practically and verify it theoretically.

$\Delta_1 = 3090 - 2350$

$\Delta_1 = 740$

$$\Delta_2 = \begin{bmatrix} 800 & 3 \\ -470 & -5 \end{bmatrix}$$

$\Delta_2 = -4000 + 1410$

$\Delta_2 = -2590$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{740}{603100} = 1.22mA$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{-2590}{603100} = -4.29mA$$

$I_1 - I_2 = 1.22 + 4.29$

$I_1 - I_2 = 5.51mA$

Marks split-up:-

S.No	DESCRIPTION	MARKS ALLOTED	MARKS obtained
1.	Experimental setup	20 marks	20
2.	observation / Readings	20 marks	18
3.	calculation/ Results	20 marks	18
4.	Viva	20 marks	18
5.	Record	20 marks	18
6.	Total	100 marks	18
			92

Result:-

Thus mesh analysis is verifyed practically and theoretically. The resultant currents for 3V / 5V supply

a) The current across 3300 resistor is 1.3mA.

b) The current across 5600 resistor is +4.15mA.

c) The current across 4700 resistor is 5.4mA.